



Methods for Co-designing Health Communication Initiatives with People with Disability: A Scoping Review

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Co-design, a collaborative approach where end-users are actively involved in design processes, has gained traction in health communication. Its integration into health communication initiatives holds promise for amplifying the voices of people with disability. Despite this, its application remains inconsistent. This scoping literature review explores how people with disability are included, when and through what methods to co-design health communication interventions. Principles of importance are considered and identified. Thirty articles published between 2018 and 2023 were reviewed. The review identified a range of methods used in health communication initiatives, with people with disability engaging at different stages and in different ways. While co-design offers potential for improved health communication, challenges such as unpredictable outcomes, diversity of participant needs, and barriers to full engagement persist. Improved transparency and consistency in reporting co-design processes are recommended to enhance rigor and effectiveness of future initiatives.

KEYWORDS

Co-design, disability, health communication, digital health, participatory action research

Co-design is recognized as a critical research and engagement approach in health communication, particularly when addressing the needs of people with disability (Fox et al., 2022). People with disability access and interface with health service systems at approximately double the rate of people without disability and are more likely to encounter barriers that impact on health service experiences (Dixon-Ibarra & Horner-Johnson, 2014; Hung et al., 2011; Kim et al., 2021; Lunsy et al., 2021; McBride Henry et al., 2023; M. A. Perry et al., 2020; Sabatello et al., 2020; World Health Organisation, 2023). Improving health services and health outcomes for people with disability requires integration of their needs and preferences into the design and delivery of services, including the ways in which information and communication interventions are designed and delivered (Brennan, 2020; Cadwgan et al., 2022; Govia et al., 2022; McBride Henry et al., 2023; World Health Organisation,

2023). Health communication initiatives, including mass media, behavior change campaigns, and digital health interventions, are critical for improving health and wellbeing of people with disability; serving as a mechanism for influencing behaviors and disseminating important messages (Rimal & Lapinski, 2009). Health communication is dynamic and often needs to be adapted to support accessibility and inclusion for people with disability (Fox et al., 2022). However, utilizing co-design in health communication is not without difficulties, particularly given the inconsistencies in how co-design is conducted, implemented, and reported within research (Chapman et al., 2024; Peters et al., 2024).

There is minimal consensus across diverse fields on methods and definitions for co-design (Peters et al., 2024). Co-design can range anywhere from consultation to more in-depth approaches where participants have influence on the research process (Peters et al., 2024). Terms like co-production and co-creation are often used interchangeably with co-design but have different definitions (Ehrlich et al., 2020). Defining what co-design entails and how it is best conducted are important for researchers and health communication practitioners, who are looking to include co-design within their practice. For the purpose of this scoping review, we have utilized a broad definition of co-design, defined as a participatory process that involves collaboration between researchers, people with disability, and other stakeholders to design health communication, which may include interventions, products, services, or other solutions (Clanchy et al., 2024; Queenslanders with Disability

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Network, 2022). Importantly, people with disability should be involved in shaping and designing the outcomes to reflect their needs (Clanchy et al., 2024; Queenslanders with Disability Network, 2022).

Despite the definitional and methodological challenges of co-designing health communication, the advantages are well documented, including: mutual respect and outcomes that reflect lived experience; equitable healthcare delivery and outcomes (Queenslanders with Disability Network, 2022; WACOSS, 2017); targeted and impactful communication strategies (Martinez Escobedo et al., n.d.); and enhanced intervention impact (Carrotte et al., 2023; Fox et al., 2022; Martinez Escobedo et al., n.d.) Co-design of health communication is particularly advantageous as it sits at the intersection of policy, services and public engagement, supporting informed decision-making (Lawless et al., 2020; Moll et al., 2020).

Despite the potential benefits of co-design of health communication with people with disability, there remains relatively little standardization, which can contribute to unclear decision-making structures, a lack of consistency in participant engagement practices, and insufficient clarity in reporting (Chapman et al., 2024). It also requires time, authentic action, transparency, and open communication – aspects still underexplored and under evaluated in academic literature (Clanchy et al., 2024; Queenslanders with Disability Network, 2022). Creating an overview of methods and building consensus on principles of importance for co-design in health communication with people with disability is beneficial to address gaps in knowledge and implementation. Understanding points of consensus and divergence in the conduct of co-design of health communication can foster consistency and better outcomes.

This scoping literature review aims to address the current gaps in health communication and co-design literature and practice by examining how people with disability are included in the co-design of health communication – including when and in what ways they are included, and what methods are used to include them. The review does not examine the effect or impact of co-design on the success of health communication initiatives, but rather how the initiatives are created and designed. This review will identify patterns, gaps, and suggestions for principles of importance to contribute to a deeper understanding of co-design and promote more inclusive, effective, and empowering health communication initiatives for people with disability.

Methods

Co-design, disability, and health communication are broad topics, vastly covered in contemporary literature together and separately. Scoping reviews are advantageous for mapping concepts, especially across broader topics or fields where different study designs and methods may be employed (Arksey & O'Malley, 2005). Unlike systematic reviews, which answer very specific research questions, a scoping review can support a more general or broader research question(s) in the pursuit of clarifying and examining the interplay of complex concepts (Arksey & O'Malley, 2005). We chose a scoping review to examine the intersection of three complex concepts, which

may require inter- and multi-disciplinary synthesis to understand how people with disability are engaged and included (Arksey & O'Malley, 2005; Bray et al., 2022; Munn et al., 2018).

As co-design is a form of inclusive research, it was important to ensure that the methods used for this review were grounded in the perspectives of lived experience of disability. The design of all aspects of the review were done collaboratively with people with disability to ensure that the results were relevant for diverse perspectives. The authorship team was formed to include a professor (EK) who lives with a degenerative disabling health condition and supports a number of family members with disability, and a research fellow (CA) with lived experience of disability. Both EK and CA supported the conceptualization, review of titles and abstracts (CA), data extraction and analysis (EK).

Review Design

The scoping review was guided by the first five stages of Arksey and O'Malley's (2005) framework for scoping literature reviews, with Levac et al. (2010) enhancements: (1) developing the research question, (2) locating relevant studies, (3) selecting studies, (4) charting the data, and (5) collating, summarizing and reporting results.

Identify Research Questions

The objective of this scoping review was to understand and describe how people with disability were included in the co-design of health communication initiatives. The review examines health communication generally, including mass media, digital health interventions, health communication applications or websites, and communication campaigns. The research questions were selected by the research team, in consultation with health communication professionals in Queensland, Australia, to address critical gaps in co-designing with people with disability. Specifically, the review sought to answer to key questions:

1. What methods are used to co-design health communication initiatives with people with disability?
2. When and how are people with disability included within the research and co-design process?
3. Is there any consensus or identification of principles of importance for co-designing health communication with people with disability?

Identify Relevant Studies

Inclusion Criteria

The studies included in this review were peer-reviewed articles and book chapters, published in English, between January 2018 and July 2023. The date range was limited to the 5 years for as co-design is an increasingly popular method with standardization occurring slowly. While there is extensive gray literature available on co-design, including co-design with people with disability, it was excluded from the scoping review for

Table 1. Inclusion and exclusion criteria for review

	Inclusion Criteria	Exclusion Criteria
Population	Adults (18+) who identify and/or are diagnosed with disability, impairment, illness or injury, including disability caused by cancer or disability caused by chronic disease.	• Children • People undergoing palliative care, end-of-life, or living with a terminal illness (besides cancer). • People living in aged care facilities. • People with chronic illness, disease, or cancer that did not identify as people with disability.
Concept	Co-design described as an intervention or method in partnership with the population.	Quantitative, qualitative or literature reviews not related specifically to co-design with the population.
Context	The design of health communication initiatives, including campaigns, marketing, messages, promotions, and resources (e.g. mobile phone applications).	All other studies not specifically related to health communication initiatives.
Type of studies	• Full-text empirical studies – quantitative, qualitative, and mixed methods that clearly articulate co-design processes. • Co-design methods papers, protocols, scoping literature reviews, and conference papers. • Peer-reviewed in academic journals, any field. • Book chapters • English • 2018–2023	Co-design scoping literature review protocols, conference abstracts, news articles. Grey literature were excluded.

pragmatic reasons (time consuming, heterogenous search strategies required). Grey literature was explored as part of the design of the research and protocol and is referenced in both the introduction and discussion. The full inclusion and exclusion criteria are shown in Table 1, based on the population, concept and context “PCC” model, which supports identifying a clear approach to scoping reviews, where the research questions and inclusion criteria are linked.

Co-design studies with children were excluded due to the additional considerations required from an ethics and consent standpoint. Similarly, co-design with people undergoing palliative care were excluded as there are additional legacy and sensitivity considerations focused on dying. While important, the focus of this review is on living with disability (Chapman et al., 2024). Furthermore, people with terminal illness do not always identify as people with disability, which is the central population to be examined by this review.

Information Sources

Six health and disability databases were searched in June and July 2023: CINAHL Complete (EBSCO), MEDLNE (EBSCO), ProQuest Central, Scopus, Taylor & Francis Online, and Web of Science.

Search Strategy

A comprehensive search strategy was developed by KC. The strategy was applied provisionally to all databases and then reviewed by CA. Medical Subject-Headings (MESH) terms were examined to create the final selection of search terms. The search was conducted by CA and KC in June and July 2023. The search strategy included abstract searching, in order to focus on the interplay of the key concepts and included the following search terms and Boolean operators:

Abstract (disab* OR impair* OR injur* or illness*) AND (codesign OR co-design) AND (health) AND (campaign OR communication OR information OR marketing OR message* OR promot* OR resource). In addition, the following terms were excluded through the use of Boolean operator NOT: (cancer OR palliative OR dying OR chronic illness OR chronic disease).

As per the inclusion and exclusion criteria, cancer and palliative care studies that do not specifically refer to people who develop disabling conditions as a result of their terminal illness were excluded, and therefore the NOT operator was used to limit their initial retrieval.

Study Selection

Two phases of screening were conducted in Covidence. First, title and abstract screening was conducted by CA with discrepancies resolved by KC. Full-text versions were screened by CA and KC. Final study selection decisions were made collaboratively and if consensus could not be reached, EK was consulted.

Charting the Data

The charting process commenced with the extraction of initial data in Covidence by CA, including paper details (lead author, publication year, title, and country) and study details (study aim, population, context, study design, co-design model/theory, co-design method/steps, outcome, and evaluation). Additional information was then extracted by CA and KC to describe how and when people with disability were included and what methods were used. The characteristics of the included studies are described in Table 2.

Table 2. Characteristics of included studies

Study details (author, year country)	Research aims and objectives	Study design, including co-design method	Study population
Adler et al. (2022), USA	To design an app for cancer survivors with disability to improve their quality of life and self-efficacy to manage cancer as a chronic condition.	Qualitative co-design and citizen science • Interviews • Workshops	Cancer survivors with disability.
Amann et al., 2020a, Switzerland	To establish a co-design approach to develop a prototype app to help people with spinal cord injury with self-management.	Qualitative co-design • Workshops • User testing	People with spinal cord injury.
Amann et al., 2020b, Switzerland	To identify benefits of a co-designed self-management app and explore factors that may impeded adoption.	Qualitative user-testing guided by Technology Acceptance Model • Semi-structured interviews	People with spinal cord injury.
Carrotte et al. (2023), Australia	To work with target audience to inform the development of a new podcast to reduce stigmatizing attitudes toward people living with complex mental health issues.	Experienced Based Co-design • Focus groups	People with complex mental health diagnoses.
Cheraghi-Sohi et al. (2023), UK	To determine the usefulness and useability of an app.	Longitudinal qualitative research and co-design • Interviews • App trial • Focus groups	People with dementia and their carers.
Davat and Martin-Juchat (2023), France	To identify patient issues and information needs relating to a new cardiac implant.	Qualitative co-design • Interviews • Workshops	People with cardiac implants.
Fortuna et al. (2021), Canada, Norway, Poland, Rwanda, USA	To describe the goals, partnerships and projects of a community of practice on patient-centered outcomes.	Longitudinal community of practice and principles of community engagement research. • Meetings and workshops • Presentations and publications • Pilot studies • Quantitative assessments	People with serious mental illness.
Fox et al. (2022), UK	To explore co-design benefits when developing smartphone apps.	Integrated patient co-design • Workshops	People with dementia and their carers.
Henni et al. (2022), Norway	To describe the needs and barriers of people using digital health solutions.	Participatory design strategies and scoping literature review • Interviews • User-testing • Evaluation	People with cognitive, mental, motor, visual, and hearing impairments that affect their use of digital health solutions.
Irvine et al. (2023), UK	To co-design a text message intervention to promote physical activity and provide support for people with stroke after they finish formal rehabilitation.	UK Medical Research Council Framework for Complex Interventions and iterative co-design • Workshops • Survey • Interviews	People with lived experience of stroke.
Jones et al., 2020, UK	To evaluate initial acceptability of a physical activity promotion tool that was co-designed with people with Huntington's disease	Qualitative co-design • Workshop	People with Huntington's Disease.

(Continued)

Table2. (Continued)

Study details (author, year country)	Research aims and objectives	Study design, including co-design method	Study population
Kerin et al. (2020), Ireland	To develop recommendations for improvement of the transition to adult health services for women with 22q11.2ds.	Participatory action research using creative narrative methods • Workshops	Women with 22q11.2ds (a rare genetic condition).
Klein et al. (2019), Australia	To explore whether an app for quitting smoking could prevent relapse.	Australian Experienced Based Co-Design • Interviews	People with severe mental illness.
Lawn et al. (2018), Australia	Describe a protocol to adapt, pilot test and evaluate a smoking cessation intervention.	Qualitative co-design and action research principles • Interview • Focus groups • Pilot trials	People with severe mental illness.
Layton et al. (2022), Australia	To reflect on co-design and inclusive research practices used to create allied health resources.	Disability Inclusive Research Principles • Meetings • Workshops	People with lived experience of disability.
Leorin et al. (2019), Italy	To co-design personalized mHealth technologies.	Sanders and Stappers Co-design model and participatory design principles. • Workshops	People with dementia.
Loton & Neal, 2020, Australia	To co-design a social marketing campaign to reduce help seeking stigma for people experiencing crisis.	Qualitative co-design • Needs assessment • Workshops • Interviews • Survey	People experiencing crises related to mental health, alcohol and other drugs.
Marier-Deschenes et al., 2021, Canada	To describe the needs and expectations of people with traumatic brain injury when co-designing an information toolkit about post-TBI sexuality.	Iterative user-centered co-design • Focus groups • Interviews	People with traumatic brain injury.
Mehmet et al. (2020), Australia	To describe a social marketing and digital communication co-design initiative.	Participatory action research and qualitative co-design • Focus groups • Questionnaires	People with mental illness.
Molinari-Ulate et al. (2023), Spain	To adapt a training and support program to co-design an online platform.	Qualitative co-design • Focus groups	People with dementia and their families.
Palmer et al. (2019), Australia	To develop a theoretical model of change to evaluate future mental health experience co-design trials and improve understanding and examination of co-design.	Experienced Based Co-Design • Training • Meetings and workshops	People experiencing severe mental illness.
M. Perry et al. (2022), New Zealand	To co-design a culturally responsive group-based pain management program.	Modified participatory action research and qualitative co-design • Meetings • Focus groups • Beta testing	Māori people living with persistent pain and Māori community members.
Rajapakse et al. (2018), Australia	To describe a co-design processes to develop artifacts (e.g., user profiles and video stories) to support communication.	Qualitative co-design • Meetings • Workshops	People with intellectual disability.

(Continued)

Table2. (Continued)

Study details (author, year country)	Research aims and objectives	Study design, including co-design method	Study population
Sin et al. (2019), UK	To integrate participatory research methodologies into the eHealth design and build process to improve usability and acceptability.	UK National Institute for Health Research co-design cycle and participatory research • Workshops	People with lived experience of psychosis and their carers.
Strudwick et al. (2021), Canada	To develop a web-based resource to help mental health providers and patients engage in the use of digital mental health tools.	Word Café • Workshop	People with lived experience of mental illness.
Tapsell et al. (2020), Australia	To describe how people with mental illness collaborated on a resource designed to highlight strategies for maintaining wellbeing.	Participatory action research • Meetings • Survey • Focus Group	People with lived experience of mental illness.
Terrill et al. (2019), USA	To describe the collaborative process used to build a web-based app for people with spinal cord injury.	Qualitative co-design • Focus groups • Feedback • User testing	People with spinal cord injury
Vasilica et al. (2020), UK	To describe the creation of an online network of social media platforms.	User-centered design and longitudinal co-design /mixed methods. • Social media engagements and measures • Events • Evaluations	People with chronic kidney disease and specialist professionals.
Vial et al. (2022), Canada	To understand how human-centered design is included in research on e-mental health interventions.	Literature review and qualitative co-design.	People with lived experience of mental illness.
Yates et al. (2023), Australia	To co-design digital, peer-delivered interventions.	Qualitative co-design as part of larger longitudinal study • Survey	Patients with chronic illness

Collating, Summarizing, and Reporting the Results

Both narrative synthesis methods (Barnett-Page & Thomas, 2009) and descriptive qualitative content analysis (Munn et al., 2018) were used to examine the extracted data. Principles of importance and discussion of key approaches were themed by KC and consensus between included studies were noted, although consensus does not explicitly represent best practice. Two researchers coded (KC and CA) and consensus was reached through discussion.

Results

Descriptive Results

The database searches identified a total of 965 records. After removing 236 duplicates, 728 titles, and abstracts were screened and 677 were removed because they used

the incorrect intervention (not co-design) or incorrect population (not people with disability). A small number ($n = 35$) were irrelevant. Fifty-one full-texts were sought for retrieval with 21 excluded, including 14 which were the incorrect intervention and 6 that examined the incorrect population. Thus, the review yielded 30 articles representing 28 discrete studies, as described in Figure 1.

Included articles were mostly conducted in Australia ($n = 10$) (Carrotte et al., 2023; Klein et al., 2019; Lawn et al., 2018; Layton et al., 2022; Loton & Neal, 2021; Mehmet et al., 2020; Palmer et al., 2019; Rajapakse et al., 2018; Tapsell et al., 2020; Yates et al., 2023) and Canada ($n = 3$) (Marier-Deschênes et al., 2021; Strudwick et al., 2021; Vial et al., 2022). Notably, a single study was conducted across multiple countries, including Rwanda (Fortuna et al., 2021). The included articles targeted diverse groups of people with disability, as described in Table 3, while the context and concepts are described below.

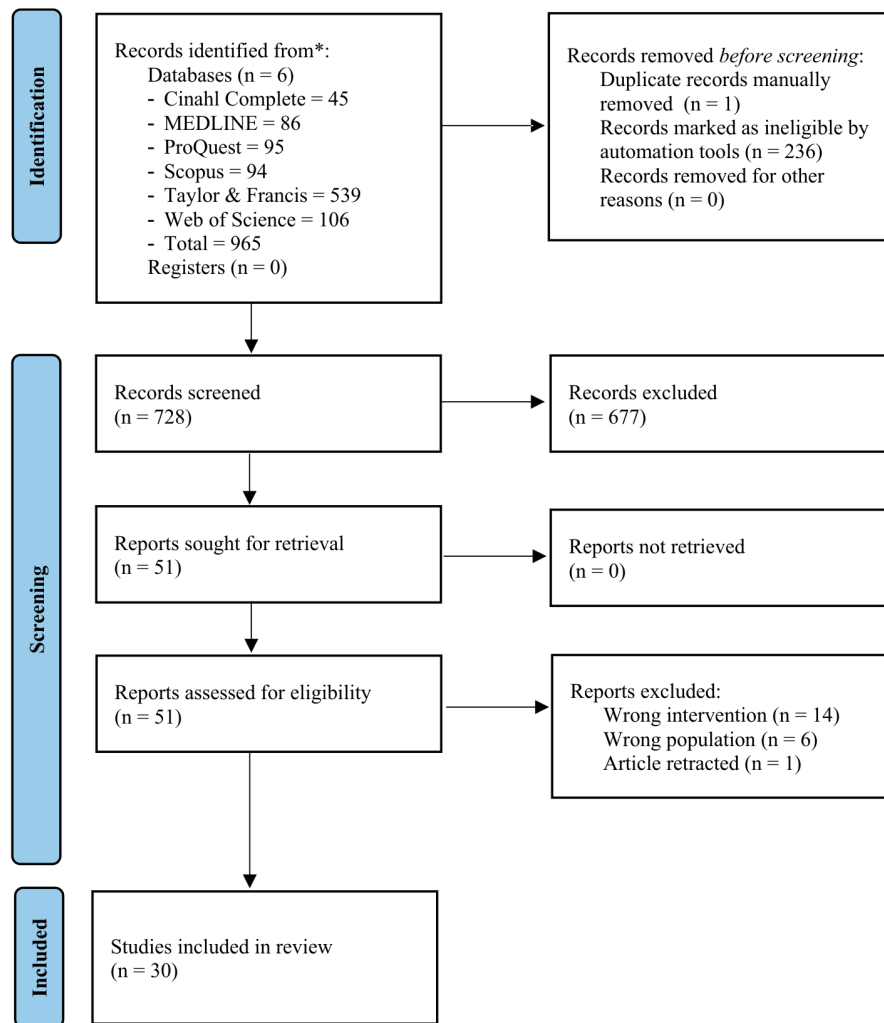


Figure 1. Flow diagram for review (adapted from PRISMA).

Table 3. Description of population in included studies

Population description	Included Articles
People with mental illness ($n = 10$)	Carrotte et al. (2023); Fortuna et al. (2021); Kerin et al. (2020); Lawn et al. (2018); Loton and Neal (2021); Mehmet et al. (2020); Palmer et al. (2019); Sin et al. (2019); Strudwick et al. (2021); Tapsell et al. (2020)
People with cognitive and communication impairments (including dementia and intellectual disability) ($n = 5$)	Cheraghi-Sohi et al. (2023); Leorin et al. (2019); Molinari-Ulate et al. (2023); Rajapakse et al. (2018)
People with spinal cord injuries ($n = 5$)	Amann, Fiordelli, Brach, et al., 2020a; Amann, Fiordelli, Scheel-Sailer, et al., 2020b; Terrill et al. (2019)
People experiencing disabling and persistent pain ($n = 2$)	M. Perry et al. (2022); Yates et al. (2023)
People with chronic kidney disease ($n = 1$)	Vasilica et al. (2020)
People experiencing disabling heart failure ($n = 1$)	Davat and Martin-Juchat (2023)
People who experienced stroke ($n = 1$)	Irvine et al. (2023)
People with Huntington's disease ($n = 1$)	Jones et al. (2021)
People with traumatic brain injuries ($n = 1$)	Marier-Deschênes et al. (2021)
People with a genetic disorder called 22q11.2ds ($n = 1$)	Kerin et al. (2020)

Context of Included Studies

While the overarching context for the included studies was the design of health communication, the aim and type of health communication was diverse. Fourteen studies focused on co-design of digital communication (e.g. mobile applications, tools, or text message interventions) to support people managing their disability (Adler et al., 2022; Amann, Fiordelli, Brach, et al., 2020a; Amann, Fiordelli, Scheel-Sailer, et al., 2020b; Cheraghi-Sohi et al., 2023; Fox et al., 2022; Leorin et al., 2019; Molinari-Ulate et al., 2023; M. Perry et al., 2022; Rajapakse et al., 2018; Strudwick et al., 2021; Tapsell et al., 2020; Terrill et al., 2019; Vasilica et al., 2020; Yates et al., 2023). Several studies co-designed communication to target specific behavior, including increasing physical activity ($n=2$) (Irvine et al., 2023; Jones et al., 2021), smoking cessation ($n=2$) (Klein et al., 2019; Lawn et al., 2018), and understanding sexuality post-injury ($n=1$) (Marier-Deschênes et al., 2021). Additionally, individual studies focused on co-designing initiatives to improve patient information (Davat & Martin-Juchat, 2023) and patient-clinician communication (Kerin et al., 2020). Three included articles reflected on longer term programs that included health communication as part of a larger initiative, including peer support and recovery ($n=1$) (Fortuna et al., 2021), mental health experience ($n=1$) (Palmer et al., 2019), and inclusive digital information ($n=1$) (Layton et al., 2022). Two reviewed the co-design of digital health solutions (Henni et al., 2022; Vial et al., 2022), while the remaining three focused on co-designing social marketing initiatives for mental health conditions (Carrotte et al., 2023; Loton & Neal, 2021; Mehmet et al., 2020).

RQ 1: Methods Used to Co-design

The co-design of health communication was described and conducted in diverse ways, using a range of conceptual frameworks, and three methods: interviews, workshops/focus groups, and trials/testing.

Conceptual Frameworks

Conceptual frameworks establish the approach to co-design, including methods and underpinning assumptions for the research. Included studies most frequently used Experienced-Based Co-design, a framework designed by Australian Healthcare and Hospitals Association and the Consumers Forum of Australia ($n=4$) (Carrotte et al., 2023; Klein et al., 2019; Palmer et al., 2019). The WHO Adaptation and Implementation Guide (Molinari-Ulate et al., 2023) and the UK Medical Research Council Framework for Complex Interventions (Irvine et al., 2023) were also used. Participatory Action Research, an approach often used in inclusive research to provide power to end-users, was used by four included studies (Kerin et al., 2020; Lawn et al., 2018; M. Perry et al., 2022; Tapsell et al., 2020). Other approaches were also utilized, including participatory design ($n=4$) (Henni et al., 2022; Leorin et al., 2019; Mehmet et al., 2020; Sin et al., 2019), user-centered design ($n=2$) (Marier-Deschênes et al.,

2021; Vasilica et al., 2020), citizen science ($n=1$) (Adler et al., 2022), community of practice ($n=1$) (Fortuna et al., 2021), design box ($n=1$) (Terrill et al., 2019), and exploratory mapping review ($n=1$) (Vial et al., 2022).

Methods

Qualitative methods were used by nearly all of the studies to co-design. Workshops and focus groups were used by 25 studies in isolation or in combination with other methods, with eight using workshops only (Fox et al., 2022; Jones et al., 2021; Kerin et al., 2020; Leorin et al., 2019; Molinari-Ulate et al., 2023; Sin et al., 2019; Strudwick et al., 2021; Vial et al., 2022). Klein et al. (2019) and Amann et al. (2020b) used interviews only while Yates et al. (2023) used survey only. Surveys were used by an additional four studies in combination with other methods (Irvine et al., 2023; Loton & Neal, 2021; Mehmet et al., 2020; Tapsell et al., 2020).

Six studies used both interviews and workshops (Adler et al., 2022; Irvine et al., 2023; Lawn et al., 2018; Loton & Neal, 2021; Marier-Deschênes et al., 2021) while three studies used both workshops and involvement of people with disability in some type of pilot trial or user-testing (Amann, Fiordelli, Brach, et al., 2020a; M. Perry et al., 2022; Terrill et al., 2019). Two studies included people in dissemination processes (Fortuna et al., 2021; Vasilica et al., 2020) with eight studies including people with disability in meetings (Fortuna et al., 2021; Layton et al., 2022; Loton & Neal, 2021; Palmer et al., 2019; M. Perry et al., 2022; Rajapakse et al., 2018; Tapsell et al., 2020; Vial et al., 2022). Two studies included people with disability through interviews, workshops and user testing/trailing, the most comprehensive approaches described in the included studies (Cheraghi-Sohi et al., 2023; Lawn et al., 2018).

RQ 2: When and How are People with Disability Included

People with disability were included and engaged at different times and through different tasks, data collection activities, and roles. Five studies included people with disability early in the research process and consistently throughout (Fortuna et al., 2021; Layton et al., 2022; Palmer et al., 2019; M. Perry et al., 2022; Rajapakse et al., 2018; Sin et al., 2019; Tapsell et al., 2020). Early inclusion of people in the co-design process occurred through preliminary research of people's experience, needs and/or support prior to data collection (Adler et al., 2022; Davat & Martin-Juchat, 2023; Irvine et al., 2023; Klein et al., 2019; Loton & Neal, 2021; Yates et al., 2023). People with disability were often included in team meetings and/or as part of the research team to guide the research. However, the majority of the included studies did not integrate people with disability into the design of the research, but rather only in the co-design of the health communication initiative.

Following the co-design of the health communication initiative, 18 studies continued to engage with people with disability, the majority through user testing, piloting trials or evaluations (Amann, Fiordelli, Brach, et al., 2020; Amann, Fiordelli, Scheel-Sailer, et al., 2020b; Cheraghi-Sohi et al., 2023; Fortuna et al., 2021; Fox et al., 2022; Henni et al., 2022;

Irvine et al., 2023; Jones et al., 2021; Kerin et al., 2020; Layton et al., 2022; Loton & Neal, 2021; Marier-Deschênes et al., 2021; Mehmet et al., 2020; M. Perry et al., 2022; Tapsell et al., 2020; Terrill et al., 2019; Vasilica et al., 2020; Vial et al., 2022). Two authors mentioned plans to include people with disability in testing and evaluation in the future (Adler et al., 2022; Carrotte et al., 2023). Three studies conducted user testing and trialing of health communication that did not actually consult end-users during the design process, but only after (Fox et al., 2022; Henni et al., 2022; Jones et al., 2021). As previously mentioned, Fortuna et al. (2021) and Vasilica et al. (2020) included people with disability in the dissemination activities associated with the co-design of health communication, including opportunities to co-present findings and attend additional events.

In summary, people with disability were most often included in the research cycle during “data collection” as part of the co-design process. Some were included in the research design phase, to support the selection and/or design of research questions, aims, and methods. More than half of the studies included people with disability in additional data collection through user testing and evaluation following the completion of the co-design process, with few included in dissemination processes.

RQ3 Consensus or Identification of Principles of Importance

Neither consensus in approach nor principles of importance were identified. However, some similarities in approach and discussion of inclusive approaches did occur across the included articles, which were themed by the research team, including diverse and inclusive representation; transparent collaboration and communication; and reciprocity.

Diverse and Inclusive Representation

Diverse and inclusive representation referred to the intentional and active engagement of people with disability from varied cultural backgrounds, disability identities, and experiences. Studies that used this principle in approach or highlighted it as critical for co-design, did so in a variety of ways. Some studies used broad, open recruitment strategies (Amann, Fiordelli, Brach, et al., 2020a, 2020b; Strudwick et al., 2021) and intentional involvement of people from diverse backgrounds, professions, and positions (Molinari-Ulate et al., 2023; M. Perry et al., 2022; Sin et al., 2019; Strudwick et al., 2021) to improve the inclusivity of their approach to co-design of health communication.

People with disability were included in some capacity within either the leadership team, governance group, or research team of seven studies (Fortuna et al., 2021; Leorin et al., 2019; Mehmet et al., 2020; M. Perry A. et al.; M. Perry et al., 2022; Tapsell et al., 2020; Terrill et al., 2019; Vasilica et al., 2020). Scoping and assessing community preferences were conducted by six studies to ensure that co-design processes were tailored to unique requirements (Kerin et al., 2020; Klein et al., 2019; Leorin et al., 2019; Marier-Deschênes et al., 2021; Terrill et al., 2019; Yates et al., 2023). Importantly, three studies

intentionally included and designed with cultural considerations in mind (Loton & Neal, 2021; Molinari-Ulate et al., 2023; M. Perry et al., 2022). In a single study, consideration of cultural contexts and requirements was realized through the inclusion of a cultural advisor to provide guidance and expertise (M. Perry et al., 2022).

Transparent Collaboration and Communication

Transparent collaboration and communication referred to the open and honest exchange of ideas, information and feedback among all stakeholders, through inclusive and accessible methods. Carrotte et al. (2023) ensured that there was adequate space and time for decision-making, ensuring more equitable power dynamics for co-designers with disability. Three studies encouraged mutual learning and capacity building among stakeholders, which promoted a common context (Palmer et al., 2019; Rajapakse et al., 2018; Tapsell et al., 2020). Four studies prioritized the development of shared goals and guidelines to support transparent collaboration (Fortuna et al., 2021; Palmer et al., 2019; Rajapakse et al., 2018; Terrill et al., 2019).

The use of diverse methods of communication to reduce barriers to participation was applied by eight studies, including the use of visual cues, flexible timeframes, and universal design principles (Fox et al., 2022; Henni et al., 2022; Jones et al., 2021; Kerin et al., 2020; Layton et al., 2022; Molinari-Ulate et al., 2023; Rajapakse et al., 2018; Terrill et al., 2019). Frequent feedback loops and iteration were implemented by seven studies (Fox et al., 2022; Lawn et al., 2018; Marier-Deschênes et al., 2021; Molinari-Ulate et al., 2023; M. Perry et al., 2022; Sin et al., 2019; Terrill et al., 2019). Iteration ensured that there was adequate time and opportunity to discuss evolving needs and priorities, while maintaining transparency about progress and decisions (Irvine et al., 2023; Layton et al., 2022; Molinari-Ulate et al., 2023; M. Perry et al., 2022; Sin et al., 2019; Terrill et al., 2019).

Reciprocity

Reciprocity referred to mutual learning, cooperation and shared value for all co-design stakeholders. Reciprocity included remuneration for participation time (Adler et al., 2022; Sin et al., 2019) and travel time (Fox et al., 2022; Sin et al., 2019) to remove barriers for engagement. Reciprocity also occurred in the form of co-presentation (Fortuna et al., 2021; Vasilica et al., 2020). However, there was a lack of clarity and information across the included studies about how participants were remunerated, the rates, when and why.

Discussion

Co-design promotes the inclusion of people that research is about or will benefit in design of an outcome (McShane & Gustafsson, 2024). However, as indicated by the results, while most included studies recognized the importance of engaging people with disability in the design of health

communication, they did so in diverse and inconsistent ways, causing confusion. For example, in the included studies, inclusive representation was identified as critical and seemed to be applied more during the ideation and design stages but was infrequently reported during data analysis and dissemination, underscoring the importance of creating consistency for co-design so that other researchers and other potential participants can build shared context and understanding.

In some cases, despite best efforts, it was difficult for the authors of this review to determine whether the processes described were co-design or consultation.

Challenges of Co-design

Co-design is often easier to describe or propose than execute effectively (Queenslanders with Disability Network, 2022). Challenges like unpredictable outcomes, higher costs, greater time investments, and difficulties in standardization present hurdles for researchers and funders alike (WACOSS, 2017). Co-design regularly occurs in academic settings, where the indicators of success include high publication outputs and research funding, which directly compete with the principles and values for inclusive research, including longer timeframes and resource intensity (Greenhalgh et al., 2019; Layton et al., 2024; McShane & Gustafsson, 2024). The institutional and methodological challenges of co-design are often compounded by the lack of consistency on how to do inclusive co-design with people with disability. While there is a wealth of evidence that suggests research should be identified and led by people with disability, methods should be accessible and inclusive, and a range of research activities should be used, there are few frameworks or examples that outline how to do this at each phase of research (Chapman et al., 2022, 2024; Gustafsson, 2023; Layton et al., 2024).

What results is a focus on co-design as a method or process, often to tick a box for inclusion, while the “skills, principles and approaches necessary to authentically power share and privilege the voices of individuals and collectives has been lost” (McShane & Gustafsson, 2024). Skills required to realize inclusive co-design practices like building relationships, skilled facilitation, and developing shared context to incorporate diverse individual and collective experiences, are often under developed in both researchers and health communication professionals alike (Carrotte et al., 2023; Mehmet et al., 2020; Palmer et al., 2019). This means that people with disability often lose out in the process and outcomes are not fit-for-purpose (McShane & Gustafsson, 2024).

Improving Rigor and Consistency

While the results of this review did not reveal consensus on how to address the methodological challenges of co-design of health communications with people with disability, there were some principles of importance identified, including diverse and inclusive representation, transparent collaboration and communication, and reciprocity. These principles could be used to increase accessibility, equity, and inclusivity, but additional

research is required to validate the principles and generate evidence for how to best apply them.

Consideration of principles not included in the results of this review are also critical for improving co-design. The field of disability inclusive health research has identified key principles to consider, including, but not limited to: accessibility and inclusion, authenticity and transparency (Chapman et al., 2022; Layton et al., 2024), capacity building empowerment (Greenhalgh et al., 2019), and translation into action (Gustafsson, 2023). Co-designers and health communication professionals should draw on these principles and approaches to improve the ways in which people with disability are integrated and involved in the co-design of health communication initiatives. For example, realizing accessibility and inclusion requires researchers to consider the social, cultural, economic, education, and political dimensions in which their participants live, to minimize barriers to participation (Rioux & Valentine, 2006). Approaches to universal design in transport and infrastructure should be examined and utilized to increase accessibility and inclusion, including the flexibility, ease, and intuitive use of methods, low physical and cognitive effort required for participation, and equity in participation (Imrie & Luck, 2014; Persson et al., 2015). If the methods for co-designing are more accessible, they are less likely to exclude important viewpoints and data (; Kerin et al., 2020; Molinari-Ulate et al., 2023) and, therefore, will be more fit-for-purpose.

To start to address the challenges of co-design, while striving to improve its rigor and consistency, research funders, peer reviewers, and editorial policies must prioritize transparency in reporting. Clear reporting is essential to delineate levels of involvement, decision-making power, and methodological detail (Staniszewska et al., 2011). Researchers should report on how co-design principles are operationalized across all phases of research, from conceptualization to dissemination (Gustafsson, 2023). Frameworks like the GRIPP2 checklist, while not specifically created for co-design reporting, can offer guidance on consistent approaches to describing co-design activities, the level of involvement participants had, and the status of the impact of participant involvement (Staniszewska et al., 2011).

While the challenges of co-design are necessary to consider, they are meaningful challenges worth addressing. Although utilizing co-design processes and methods may initially require additional resources and investment of time and money, it does have the potential to significantly reduce long-term costs to systems and services by focusing on prevention, early intervention, and risk reduction (Mehmet et al., 2020; Terrill et al., 2019; WACOSS, 2017).

Conclusion

Co-design is a pivotal approach in health communication research and design. The focus on co-design signifies a shift from traditional research toward meaningful engagement and inclusion. While this review highlights the diverse ways in which co-design is approached and conducted in health communication initiatives with people with disability, it also underscores the need for greater transparency and consistency.

Despite the comprehensive approach and scope of the review, the exclusion of gray literature and selection of databases may have overlooked key insights from community-led or non-academic initiatives. The inclusion of English-language studies may also exclude culturally significant practices. The review did not assess the outcomes of the initiatives nor the overall effectiveness of co-design in health communication for people with disability. Future studies should explore both. Addressing the limitations identified will support more rigorous and inclusive co-design research moving forward.

In conclusion, while co-design holds immense promise and potential for inclusive and effective design of health communication initiatives, its successful implementation hinges on addressing existing challenges and fostering a culture of transparency. By delving deeper into the nuances of co-design processes and practices, researchers and policymakers can unlock its full potential to drive positive change in healthcare delivery and communication.

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